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Roll No

ME-5005(2)-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Metrology and Inspection

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat diagram wherever required.

1. a) Give the structure of generalized measuring system and explain. Give the difference between accuracy and precision with an example.
b) Briefly explain the modes of operation and applications of the measurement system.
2. a) Sketch a micrometer and explain its working.
b) Describe two methods of testing the parallelism of micrometer measuring faces and also discuss the limitations of each method.
3. a) Discuss several types of tolerances.
b) What do you understand by the term hole basis and shaft basis in terms of assembly fit specifications? Which is preferred? Give reason supporting your answer.

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4. a) Discuss the method of measurement of straightness.
b) Write short note on non-contact profiling system.
5. a) Discuss the factors affecting the surface roughness.
b) Describe the methods used for measurement of roughness.
6. a) Classify comparators?
b) What is 3D metrology?
7. a) Discuss thread comparators?
b) Explain working of gauge length interferometer.
8. a) Draw a diagram of a gear, pointing the following and define them:
 - i) Pitch surface ii) Pitch circle
 - iii) Clearance iv) Flank of tooth
 - v) Circular thickness vi) Root circle
- b) Explain the process of Laser Vision with its principle diagram.
